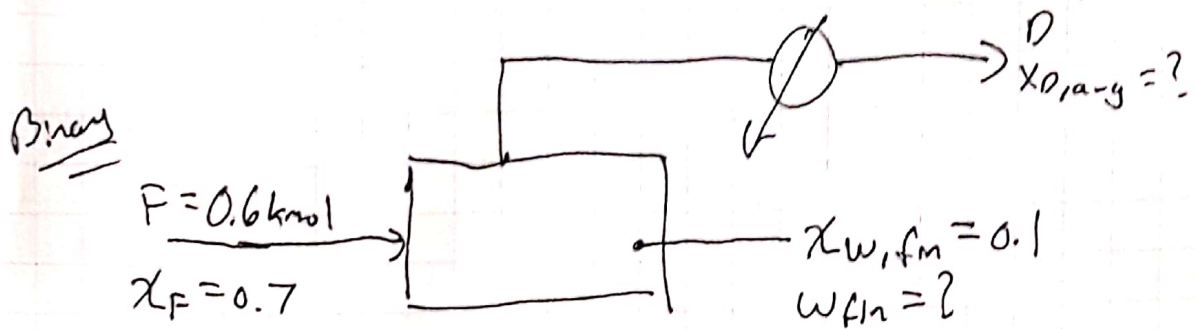


Simple batch distillation



1) Find overall mass balance & component mass balance

$$\left\{ \begin{array}{l} \text{OB: } F = D_{\text{tot}} + W_{fm} \\ \text{CMB: } Fx_F = W_{fm}x_{W, fm} + D_{\text{tot}}x_{D, avg} \end{array} \right.$$

2) Find W_{fm} , distillate D_{tot} , $x_{D, avg}$ $W_{\text{total}} = F \exp \left(- \int_{x_{\text{final}}}^{x_F} \left(\frac{dx}{y-x} \right) \right)$

x	y	$\frac{y-x}{y-x}$
0.0	0	
0.1	0.418	3.14465
0.2	0.579	2.636
0.3	0.665	2.7347
0.4	0.729	3.0395
0.5	0.779	3.589
0.6	0.825	4.449
0.7	0.870	5.8623
0.8	0.915	8.6956

$= F \exp(-\text{Area})$
 $\nearrow$
 Symptom
 Rate

$$\text{Area} = \frac{x_F - x_{\text{final}}}{6} \left[\frac{1}{y-x} \Big|_{x_{\text{final}}}^{0.1} + \frac{4}{y-x} \Big|_{x=0.4} + \frac{1}{y-x} \Big|_{x_{\text{feed}}}^{0.7} \right]$$

$$= \frac{0.7 - 0.1}{6} \left[1(3.14465) + 4(3.0395) + 1(5.8623) \right]$$

(Area = 2.1185)

$$W_{final} = F \exp(-Area)$$

$$= 0.6 \text{ kmol} \exp(-2.1185)$$

$$[W_{final} = 0.072127 \text{ kmol}] \checkmark$$

$$F = D_{tot} + W_{fin}$$

$$D_{tot} = F - W_{fin}$$

$$= 0.6 - 0.072127$$

$$[D_{tot} = 0.52787 \text{ kmol}] \checkmark$$

$$F X_F = D_{tot} X_{D,avg} + W_{fin} X_{W,fin}$$

$$D_{tot} X_{D,avg} = F X_F - W_{fin} X_{W,fin}$$

$$X_{D,avg} = \frac{F X_F - W_{fin} X_{W,fin}}{D_{tot}}$$

$$X_{D,avg} = \frac{(0.6)(0.7) - (0.072127)(0.1)}{0.52787}$$

$$[X_{D,avg} = 0.78198] \checkmark$$